Team Member: Olu **Role:** Neural Architecture & Computational Demonstration Lead

1. Introduction & Objectives

My architecture-focused role connects the safari's conceptual exhibits to the notebook's executable LeopardCNN and ParrotLSTM models. The objective is to compare temporal convolution and recurrent memory on the same synthetic sequence-classification task. This report describes the supplied implementations and recorded outputs, rather than claiming U-Net construction, diffusion scheduling, or reverse sampling.

2. Technical Execution & Methodology

2.1 Leopard: Temporal Convolution

The CNN applies Conv1d layers with 16 and 32 output channels, kernel size 5, padding 2, and stride 1. ReLU activations, max pooling, adaptive average pooling, and a 32→2 linear classifier complete the model.

2.2 Parrot: Recurrent Memory

The LSTM transposes [*B,* 1, 50] inputs to [*B,* 50, 1]. It uses one recurrent layer with input size 1 and hidden size 32; the final timestep output feeds a 32→2 classifier. No separate vanilla RNN is trained.

2.3 Shared Training Procedure

The notebook generates 1,000 sequences and uses an 800/200 train–test split, batch size 32, and 15 epochs per model. Training uses cross-entropy loss and Adam with learning rate 0.005, including backpropagation and optimizer updates.

2.4 Tensor Flow & Architecture Atlas

For the CNN, tensor shapes follow [*B,* 1, 50] → [*B,* 16, 50] → [*B,* 16, 25] → [*B,* 32, 25] → [*B,* 32, 1] → [*B,* 2]. The LSTM returns timestep features [*B,* 50, 32] before final classification. These dimensions are derived from the model definitions, not a printed shape trace.

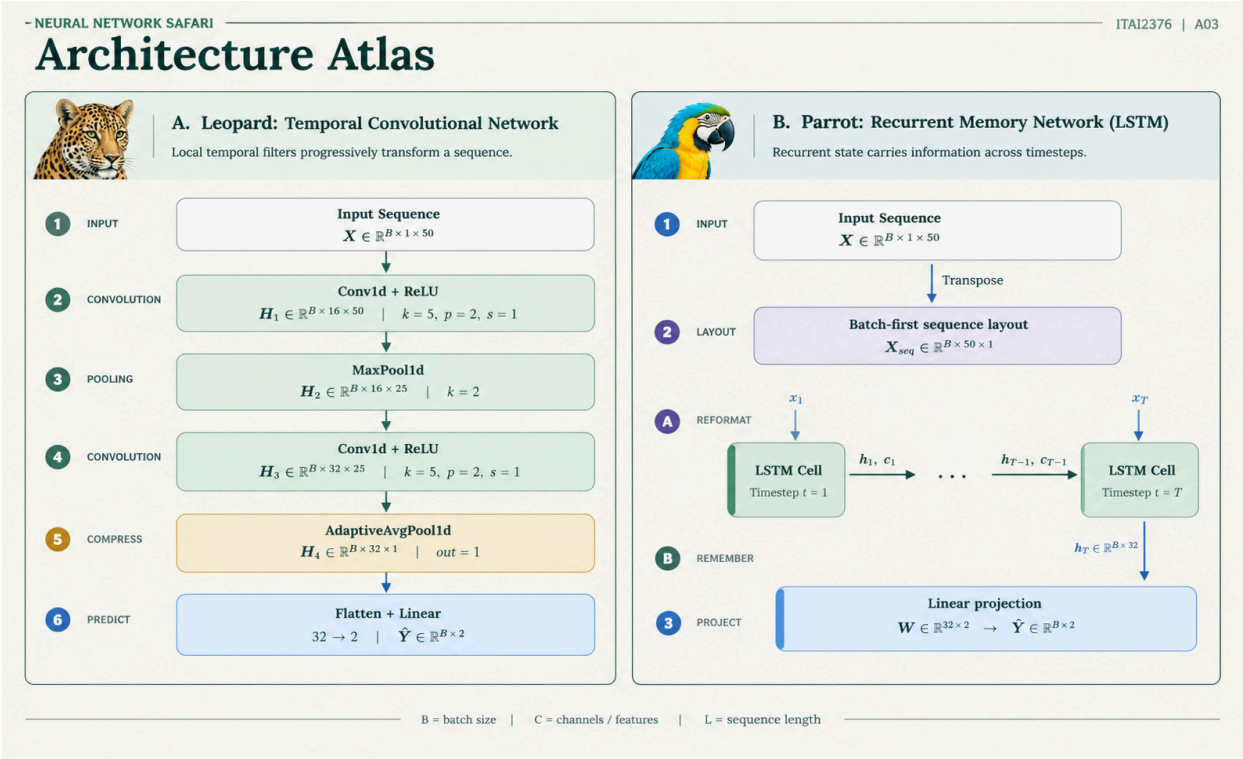


Figure 1. Supplied Architecture Atlas, reproduced as one intact image. It illustrates the CNN and LSTM tensor pathways. It is a user-supplied explanatory graphic, not an original notebook output or a training-performance measurement.

3. Recorded Results & Interpretation

Model Test accuracy

> Leopard: 1D CNN 98.00% Parrot: LSTM 48.50% Both figures come from the saved output of physical notebook cell 10 and use the 200-example test split. The difference is 49.50 percentage points.

These results describe one recorded experiment, not a general ranking of convolutional and recurrent architectures. The notebook also stores a training-loss plot, but it does not print an exact per-epoch numerical loss table. I therefore do not estimate final losses from the image or claim additional evaluation runs.

4. Challenges & Limitations

The recorded LSTM result is substantially lower than the CNN result. The notebook does not establish the cause through repeated seeds, ablation experiments, or gradient diagnostics. I therefore do not claim that normalization, residual scaling, or gradient-health unit tests resolved the performance gap.

The computational demonstration uses synthetic numerical sequences rather than animal photographs. The six safari exhibits are educational representations, but only the CNN and LSTM are implemented and trained in this demonstration. The Hyena/RNN, Ostrich/FNN, Octopus/Autoencoder, and Spider/GNN should not be described as additional trained models.

5. Reflection

This demonstration strengthened my understanding of the difference between an architectural analogy and a measured result. The Leopard model applies one-dimensional convolutions to temporal data, so the CNN concept should not be restricted to images. The Parrot represents gated memory, but an appropriate analogy does not guarantee successful learning in a particular run.

The comparison also reinforced the need to distinguish structure from training objective. In this notebook, both implemented architectures use a supervised binary-classification objective; that experimental choice should not be presented as the only objective those architectural families can support. An accurate contribution report must connect each implementation claim to visible code and each performance claim to recorded evidence.

> **Key insight**
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> Architecture determines how information moves through the model. Credible evaluation requires showing what that implementation actually achieved, including weak results and unresolved limitations.

Evidence, Image Provenance & Attribution

Technical source: supplied A03_Neural_Network_Safari_Deck(Notebook).ipynb, physical cell 10 (1-based notebook order, not execution count): synthetic data generation, LeopardCNN, ParrotLSTM, training loop, evaluation function, saved accuracy output, and loss plot.

Image source: user-supplied a jpg file that was generated through nano banana 2. The atlas is supporting artwork, not an embedded figure extracted from the notebook. Its projection matrix $W \in \mathbb{R}^{32\times2}$expresses a row-vector mapping; PyTorch stores the corresponding Linear(32,2) weight tensor with shape [2, 32]. The implementation also includes a bias term. The illustrated colorful parrot is a visual motif, not the notebook's specific African Grey identity.

Scope: the numerical results are transcribed from stored outputs; training has not been rerun for this report. The notebook supports the technical description but does not establish individual cell authorship. The member-role attribution should be confirmed before submission.